

## **Phase 1: Basic Data Structures**

### **1. Arrays**

- Operations: insert, delete, search
- Problems: Kadane's, Prefix Sum, Sliding Window

### **2. Strings**

- Reversal, Palindrome check
- Pattern Matching: KMP, Rabin-Karp

### **3. Linked Lists**

- Singly, Doubly, Circular
- Fast and slow pointers (Floyd's Cycle)
- Reversal, Merge two sorted lists

### **4. Stacks and Queues**

- Stack (infix to postfix, balanced parentheses)
- Queue, Circular Queue
- Deque
- Monotonic Stack/Queue

### **5. Hashing**

- Hash Maps (C++: `unordered_map`)
- Applications: frequency count, pair sum problems

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## **Phase 2: Recursion & Backtracking**

- Factorial, Fibonacci
  - Permutations, Combinations
  - N-Queens, Sudoku Solver
  - Subset Sum, Rat in a Maze
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### Phase 3: Sorting and Searching

- Sorting Algorithms:
    - Bubble, Selection, Insertion
    - Merge Sort, Quick Sort, Heap Sort
  - Searching Algorithms:
    - Binary Search (standard + variations)
    - Search in Rotated Sorted Array
  - Two Pointer Technique
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### Phase 4: Intermediate Data Structures

#### 1. Binary Trees

- Pre/In/Post Order
- Height, Diameter, Balanced Check
- Lowest Common Ancestor (LCA)

#### 2. Binary Search Trees (BST)

- Insert/Delete/Search
- Convert BST to Balanced BST
- Kth Smallest/Largest Element

#### 3. Heaps

- Min Heap, Max Heap
- Priority Queue (C++: `priority_queue`)
- Heapify, Top-K problems

#### 4. Tries

- Insert/Search/Prefix Matching
- Word Dictionary, Autocomplete

#### 5. HashSet/HashMap (Applications)

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## Phase 5: Core Algorithmic Techniques

### Dynamic Programming (DP)

- Fibonacci, Climbing Stairs
- 0/1 Knapsack, Subset Sum
- Longest Increasing Subsequence
- Longest Common Subsequence
- Matrix Chain Multiplication
- DP on Trees
- Bitmask DP
- DP with Binary Search

### Greedy Algorithms

- Activity Selection
- Fractional Knapsack
- Huffman Encoding
- Job Scheduling

### Sliding Window

- Fixed/Variable Window
- Max/Min in sliding window
- Anagrams in a String

### Divide and Conquer

- Merge Sort
- Count Inversions
- Closest Pair of Points

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## Phase 6: Advanced Data Structures

- **Segment Trees**
    - Range Sum, Range Min
    - Lazy Propagation
  - **Fenwick Tree (Binary Indexed Tree)**
  - **Disjoint Set (Union-Find)**
    - Path compression, Union by Rank
    - Applications: Kruskal's algorithm, Connected components
  - **AVL Tree, Red-Black Tree (Theory only for interviews)**
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## **Phase 7: Graph Algorithms**

- Representation (Adjacency list/matrix)
  - BFS, DFS
  - Topological Sort
  - Dijkstra's Algorithm
  - Bellman-Ford
  - Floyd-Warshall
  - Minimum Spanning Tree: Prim's, Kruskal's
  - Detect Cycle (Directed & Undirected)
  - Kosaraju's Algorithm (SCC)
  - Bipartite Graph, Bridges, Articulation Points
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## **Phase 8: Miscellaneous Concepts**

- Bit Manipulation
- Mathematical Algorithms
  - GCD, LCM, Prime Sieve (Eratosthenes)
  - Modular Arithmetic, Fermat's Little Theorem

- Sliding Window Maximum
  - Two/Three Pointer Problems
  - Binary Search on Answer
  - Meet in the Middle
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